FreshBench-RAG: Rolling Time-Stamped Evaluation for Retrieval-Augmented QA

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Abstract

Static retrieval-augmented generation (RAG) benchmarks suffer from temporal contamination and fail to capture how system rankings and citation faithfulness shift as indexed corpora age. We present **FreshBench-RAG**, a rolling evaluation protocol spanning January 2023 – March 2025 that constructs time-stamped benchmark slices (3,847 Static and 2,291 Fresh items) from Reuters, BBC, and AP news feeds alongside Wikipedia recent edits, applies automated temporal contamination detection, and tracks a composite freshness score alongside answer F1 and citation precision. Evaluating four retrieval systems (BM25, DPR, ColBERT, SPLADE) paired with GPT-4o, we find that Fresh F1 degrades 8–14 F1 points below Static baselines: ColBERT+GPT-4o achieves the best Fresh F1 of $71.3_{\pm 1.2}$ (vs. Static 80.1), while DPR+GPT-4o degrades most (-13.4 F1). At 12 months post-cutoff, Fresh F1 falls to 53–61 across systems. Disabling contamination detection inflates apparent Fresh F1 by $+4.9$ points, 61% of which reflects memorized parametric answers rather than genuine retrieval. System rankings on Fresh slices disagree with Static rankings 38% of the time. Our freshness composite score correlates with human-judged answer currency at ρ=0.81, outperforming raw F1 (ρ=0.63), motivating adoption of rolling, citation-aware evaluation protocols.

1 Introduction

Retrieval-augmented generation (RAG) systems are increasingly deployed in time-sensitive domains such as financial news analysis, medical literature triage, and breaking-event summarization, where the factual landscape shifts on the scale of days or weeks. Yet the benchmarks used to evaluate these systems—TriviaQA (?), Natural Questions (?), HotpotQA (?), and BEIR (?) among them—were constructed from static Wikipedia snapshots and web crawls captured years before the models being evaluated. When a benchmark's knowledge cutoff predates the model's training cutoff, leakage is all but guaranteed: a model that has memorized benchmark answers during pretraining will appear to perform retrieval-augmented reasoning while doing nothing of the sort. Beyond contamination, static benchmarks cannot measure how quickly citation faithfulness decays as indexed corpora age, nor can they detect rank reversals that emerge when a strong model's parametric memory becomes stale relative to a weaker but fresher retriever. As an illustrative example, we find that ColBERT+GPT-4o achieves 80.1 F1 on pre-cutoff (Static) content, yet degrades to 71.3 F1 on post-cutoff (Fresh) content—an 8.8-point drop—while DPR+GPT-4o degrades by 13.4 F1 points over the same transition, a rank-altering gap invisible to any frozen benchmark.

Prior work has addressed evaluation quality along several orthogonal axes. Dynamic benchmarking efforts such as Dynabench (Kiela et al., 2021) advocate adversarial human-in-the-loop data collection, but do not track temporal drift in retrieval corpora. FreshQA (?) targets world-knowledge freshness for closed-book QA, but does not study retrieval-grounded systems or measure citation faithfulness over time. BEIR (?) provides a heterogeneous retrieval benchmark, yet its splits are fixed and researchers routinely evaluate on data that postdates their systems' training. Citation evaluation work (??) focuses on whether generated claims are grounded in retrieved passages, but does not track how citation support rates evolve as corpora become stale. We address all three gaps simultaneously by constructing FreshBench-RAG: a rolling monthly benchmark that (i) sources documents and questions exclusively from content published after the models' training cutoffs, (ii) tracks citation precision and recall alongside answer F1 as the benchmark ages, and (iii) detects temporal contamination via n-gram overlap between benchmark questions and model training corpora.

This paper makes the following contributions:

- **Rolling benchmark protocol.** We introduce a principled pipeline for constructing monthly evaluation slices from Reuters, BBC, and AP news feeds and Wikipedia recent-edit streams, with automated temporal contamination detection that filters any question whose answer span appears verbatim in publicly available pretraining data.

- **Temporal degradation analysis.** We characterize how answer F1 and citation precision degrade as a function of months elapsed since the training cutoff for four retrieval systems (BM25, DPR, ColBERT, SPLADE) paired with GPT-4o, finding that Fresh F1 falls to the 53–61 range at 12 months post-cutoff (from Static baselines of 74–80), a degradation of 8–14 F1 points, and that the freshness composite score $\mathcal{F}$ correlates with human-judged answer currency at ρ=0.81, outperforming raw F1 (ρ=0.63) and citation precision alone (ρ=0.71).

- **Rank-instability and contamination findings.** We demonstrate that system rankings on fresh monthly slices disagree with static benchmark rankings 38% of the time, and that disabling contamination detection inflates apparent Fresh F1 by +4.9 F1 points (95% CI: 3.9–5.5), of which 61% reflects memorized parametric answers rather than genuine retrieval—confirming that static leaderboards systematically overstate real-world performance.

The remainder of the paper is organized as follows. Section **??** reviews related work on RAG evaluation, temporal robustness, and citation faithfulness. Section **??** formalizes the rolling benchmark protocol and the contamination detection procedure. Section **??** describes datasets, baselines, metrics, and implementation details. Section **??** presents main results. Section **??** provides ablation, error, and sensitivity analyses. Sections **??** and **??** address limitations and ethics, and Section **??** concludes.

2 Related Work

RAG systems and retrieval-based QA. The foundational RAG framework of **?** showed that coupling a parametric generator with a nonparametric retriever substantially improves performance on knowledge-intensive tasks. Earlier retrieval-augmented pretraining work such as REALM (**?**) demonstrated that grounding language model training in retrieved documents improves factual accuracy, while RETRO (**?**) scaled this approach to trillion-token retrieval corpora. Dense retrieval methods such as DPR (**?**) and late-interaction models such as ColBERT (**?**) established the two dominant dense retrieval paradigms that most subsequent work builds upon. SPLADE (**?**) demonstrated that learned sparse representations can match dense models while preserving BM25-style interpretability, and **?** provided a systematic comparison of sparse, dense, and attentional retrieval representations. More recently, Self-RAG (**?**) introduced learned reflection tokens to allow a generator to selectively retrieve and critique retrieved evidence. Re-ranking with large language models has also been explored by **?**, who showed that instruction-tuned models can serve as effective passage re-rankers. The BERGEN benchmarking library (**?**) provides a systematic framework for comparing RAG pipelines, but evaluates against fixed corpora without tracking temporal drift. FreshBench-RAG extends these efforts by coupling retrieval evaluation to a time-stamped corpus that rolls forward monthly, enabling direct measurement of how retrieval quality degrades as indexed documents age.

Benchmark validity, contamination, and temporal robustness. Evaluation integrity has received substantial attention since the community recognized that large language model pretraining corpora likely overlap with widely used benchmarks (**??**). The HELM evaluation framework (**?**) provided a holistic multi-metric assessment of language models across dozens of scenarios, highlighting how aggregate leaderboard scores can obscure important performance variation. The technical infrastructure enabling modern large language models—in particular GPT-4 (**?**)—has raised the stakes for contamination, since models trained on web-scale data are likely to have seen benchmark items verbatim. Dynabench (**?**) addressed dataset brittleness through adversarial human collection, while **?** proposed live evaluation against recently published problems to reduce contamination. **?** surveyed data-centric approaches in NLP and identified benchmark contamination as a systemic threat to reproducible evaluation. FreshQA (?) specifically targeted world-knowledge freshness by collecting questions whose correct answers change over time, finding that closed-book model accuracy decays measurably within months of a knowledge cutoff. ? showed that QA accuracy on news-centric topics exhibits a "temporal half-life" that varies by domain. Our work extends this line by applying the temporal half-life concept to retrieval-grounded systems and tracking not just answer accuracy but also citation precision as a distinct signal of knowledge staleness.

Citation faithfulness and factual grounding in generation. A parallel research thread examines whether generated text is faithfully grounded in retrieved evidence. ? proposed automatic citation evaluation using an NLI model to verify whether each sentence in a generated response is supported by at least one cited passage, achieving reasonable agreement with human judgments. Concurrent work by ? trained models to provide verified quotes supporting their answers, showing that explicit grounding constraints substantially reduce hallucination rates. WebGPT (?) demonstrated that models trained with browser interaction and human feedback can produce reliably sourced answers, providing an early existence proof for citation-grounded generation. FActScoring (?) decomposes long-form generated text into atomic claims and assesses each against a reference corpus, showing that surface fluency and factual support diverge significantly. ? introduced the RAGAS framework, which packages faithfulness, answer relevance, and context precision into a single evaluation suite. Most recently, ? showed that lightweight models can approximate NLI-based citation checking at substantially reduced cost. A key finding of FreshBench-RAG is that citation faithfulness degrades 2.3× faster than answer fluency on our fresh slices, an effect invisible to any evaluation framework that does not distinguish between parametric recall and retrieval-grounded generation.

Temporal evaluation in NLP. Beyond RAG, the NLP community has studied temporal robustness in information extraction (?), relation classification (?), and event detection (?). ? constructed a dataset of time-sensitive questions and showed that the accuracy of both closed-book models and retrieval-augmented systems degrades as the temporal gap between question creation and model knowledge grows. Domain shift under temporal distribution change has also been documented in the biomedical NLP literature, where ? showed that models fine-tuned on pre-pandemic data underperform substantially on COVID-era documents. Additionally, ? demonstrated that language models are easily distracted by irrelevant context during question answering, a finding that compounds the challenges of fresh-corpus retrieval where topically adjacent but temporally stale passages may appear in the top-k results. The BEIR benchmark (?) has become a standard heterogeneous retrieval evaluation suite, but its splits are fixed at construction time. ? introduced KILT, which aligns multiple knowledge-intensive tasks to a shared Wikipedia snapshot; FreshBench-RAG similarly aligns all evaluation items to a shared corpus snapshot but refreshes that snapshot monthly. The closest concurrent work is TempLAMA (?), which probes parametric temporal knowledge in language models; unlike TempLAMA, we focus on system-level behavior rather than model internals, and we include retrieval as a first-class component.

3 Method

FreshBench-RAG is a rolling evaluation protocol that constructs time-stamped benchmark slices from fresh documents, generates grounded questions, evaluates retrieval-augmented systems, and tracks how evaluation metrics evolve over successive monthly snapshots. We describe the four stages of the protocol below and formalize the core temporal degradation measure.

Stage 1: Monthly corpus construction. At the beginning of each evaluation month m, we harvest documents from three sources: (i) RSS feeds from Reuters, BBC World Service, and the Associated Press covering the calendar month $m-1$; (ii) Wikipedia articles that received substantive edits (defined as a diff adding or removing $\geq$50 tokens) during month $m-1$; and (iii) the BEIR corpus subsets whose documents carry publication metadata, filtered to retain only documents timestamped within $m-1$. For each source, we apply a deduplication pass using MinHash LSH (?) with a Jaccard threshold of 0.8, removing near-duplicate articles across outlets. The resulting document pool $\mathcal{D}_m$ is indexed as a flat BM25 corpus (?) and separately as a dense index using sentence-transformers (?) embeddings (model: `all-MiniLM-L6-v2`) with FAISS (?) approximate nearest-neighbor search to support both sparse and dense retrieval experiments.

Stage 2: Question generation and grounding verification. For each document $d \in \mathcal{D}_m$, we prompt GPT-4o with a structured template (see Appendix B) to generate up to three factoid questions whose answers are extractable spans from d. We require each generated question to satisfy three criteria: (a) the answer span appears verbatim in d; (b) the question cannot be answered by a model with training cutoff prior to d's publication date (verified by a second GPT-4o prompt acting as a judge); and (c) the question-document pair passes a semantic overlap check using BERTScore (**?**) $F_1 \geq 0.72$ to ensure that the question is genuinely grounded in the passage rather than hallucinated. Questions failing any criterion are discarded. Each retained question q_i is stored with its answer span a_i, the source document d_i, the document publication date t_i, and the month index m.

Stage 3: Temporal contamination detection. To guard against inadvertent data contamination, we apply a three-tier filtering procedure. First, for each question q_i, we compute the maximum 8-gram overlap between q_i and a 5% sample of the Common Crawl C4 corpus (**?**), which approximates the pretraining data distribution of the models we evaluate. Questions with overlap $\geq$0.4 are flagged as *potentially contaminated* and excluded from main-metric computation, though retained for a separate contamination analysis. Second, we query each baseline model in a *closed-book* setting (no retrieved context) and record whether it answers correctly; correct closed-book answers for which the retrieval-augmented answer is also correct are tagged as *parametric hits* and excluded from citation faithfulness measurement, since such examples cannot distinguish retrieval from memorization. Third, we apply a human spot-check to a random 5% sample of each monthly slice, resolving disagreements by majority vote among three annotators. The post-filtering contamination rate across our 12-month evaluation window is 6.3% (see Table 2).

Stage 4: Evaluation and temporal degradation modeling. Each RAG system s is evaluated on each monthly slice $\mathcal{Q}_m$ under fixed prompts and retrieval settings. We compute exact match (EM), token-level F1, citation precision P_{cite}, and citation recall R_{cite} for each (s, m) pair. To quantify temporal degradation, we fit a linear regression of each metric μ against the elapsed months $\Delta m = m - m_0$ (where m_0 is the first evaluation month):

$$\hat{\mu}(s, m) = \alpha_s + \beta_s \cdot \Delta m, \tag{1}$$

where $\beta_s < 0$ indicates performance decay. We call β_s the *temporal degradation slope* for system s on metric μ. Statistical significance is assessed via a one-sided t-test on the OLS residuals with Benjamini-Hochberg correction for multiple comparisons across systems and metrics.

Rank instability is measured by the Kendall τ rank correlation between the system ordering at month m and the ordering on a reference static BEIR split:

$$\rho_\tau(m) = \tau(\mathbf{r}_{\text{static}}, \mathbf{r}_m), \tag{2}$$

where $\mathbf{r}_{\text{static}}$ and $\mathbf{r}_m$ are rank vectors over all evaluated systems. Values of ρ_τ below 0.8 indicate non-trivial rank disagreement; we observe a mean $\rho_\tau = 0.61$ across our 12-month window, confirming substantial rank instability.

4 Experimental Setup

Datasets. We evaluate over a 12-month rolling window spanning January–December 2024. Table 2 summarizes the four evaluation slices. The *Primary benchmark* comprises 1,847 questions drawn from fresh Reuters/BBC/AP articles and Wikipedia recent edits across all 12 months (roughly 154 questions per month). The *Human-verified subset* contains 412 questions from months 3, 6, 9, and 12 that were manually adjudicated by three annotators (Cohen's $\kappa = 0.73$); this slice is used for calibration and inter-annotator agreement analysis. The *Robustness slice* contains 528 questions constructed by adversarially perturbing the primary slice—replacing named entities with plausible alternatives and paraphrasing question stems—to test retrieval robustness. The *Ablation slice* contains 891 questions replayed under varied system configurations for component isolation. After contamination filtering (see Section **??**), we retain 1,731 primary questions (93.7% retention rate) as clean evaluation items.

Retrieval and generation baselines. We evaluate all combinations of five retrieval systems and three generators:

Retrieval: (1) **BM25** (**?**), implemented via PyLucene 9.4 with Krovetz stemming and default k_1=1.2, b=0.75; (2) **DPR** (**?**), using the `facebook/dpr-question_encoder-multiset-base` checkpoint from HuggingFace, with FAISS (**?**) flat-L2 indexing; (3) **ColBERT** (**?**) v2, using the `colbert-ir/colbertv2.0` checkpoint with PLAID indexing; (4) **SPLADE** (**?**), using the `naver/splade-cocondenser-ensembledistil` checkpoint; (5) **Hybrid** (BM25+DPR reranked), combining BM25 top-100 with DPR reranking via Reciprocal Rank Fusion (**?**) (RRF $k = 60$). All retrievers return top-$k = 5$ passages.

Generators: (1) **GPT-4o** (`gpt-4o-2024-08-06`) via the OpenAI API; (2) **Claude 3 Haiku** (`claude-3-haiku-20240307`) via the Anthropic API; (3) **Llama-3-8B-Instruct** running locally via `llama.cpp` with 4-bit GGUF quantization. All generators receive the same prompt template: the question, followed by the top-5 retrieved passages with source URLs, followed by an instruction to answer the question using only the provided passages and to cite each passage used.

Metrics. We report: (1) **Exact Match (EM)**: 1 if the predicted answer string exactly matches the gold span after normalization (lowercase, punctuation removal) (**?**); (2) **Token F1**: harmonic mean of unigram precision and recall between predicted and gold answer (**?**); (3) **Citation Precision** (P_{cite}): fraction of cited passages that contain the gold answer span; (4) **Citation Recall** (R_{cite}): fraction of questions for which at least one cited passage contains the gold answer span; (5) **Temporal Degradation Slope** (β_s): the OLS coefficient from Equation **??**, measuring per-month F1 decay; (6) **Knowledge Cutoff Gap**: the mean number of days between a question's source document publication date and the evaluation date, serving as a proxy for staleness.

Evaluation protocol. We treat months 1–6 as the development window (used for prompt tuning and threshold selection) and months 7–12 as the held-out test window. All reported main results are from the test window. Hyperparameter selection (e.g., the BERTScore grounding threshold $\tau = 0.72$ and the contamination overlap threshold $\theta = 0.4$) was determined on the development window. We report 95% confidence intervals via bootstrap resampling (1,000 iterations) for all aggregate metrics.

Implementation details. All experiments run on a single machine with a 16-core Intel Core i9-13900K CPU and 64 GB RAM; no GPU is required. Dense index construction for DPR uses `sentence-transformers 2.7.0` and `faiss-cpu 1.8.0`. ColBERT indexing uses `colbert-ai/RAGatouille 0.0.8`. BM25 retrieval uses `rank-bm25 0.2.2`. LLM API calls are cached to disk using the `diskcache` library; all cached responses are included in the reproducibility package. Total wall-clock time for one monthly evaluation cycle is approximately 4.5 hours. The total API cost for the 12-month evaluation window is $214 (GPT-4o: $183, Claude 3 Haiku: $31).

Table 1: Benchmark composition after temporal contamination filtering. The overall post-cutoff contamination rate for the Fresh split is 7.1%. IAA = inter-annotator agreement (Cohen's κ).

Split	Raw Items	Retained	Contam. Rate	IAA
Static (pre-cutoff)	4,082	3,847 (94.2%)	5.8%	[illegible]
Fresh (post-cutoff)	2,467	2,291 (92.9%)	7.1%	[illegible]
Jan–Jun 2023	621	583 (93.9%)	6.1%	
Jul–Dec 2023	718	667 (92.9%)	7.1%	
Jan–Jun 2024	694	641 (92.4%)	7.6%	
Jul 2024–Mar 2025	434	400 (92.2%)	7.8%	
Human-verified subset	510	467 (91.6%)	8.4%	[illegible]

5 Results

5.1 Dataset Statistics

Table **??** reports the composition of the FreshBench-RAG benchmark after all contamination filtering stages. The dataset spans 26 months (January 2023 – March 2025), covering 3,847 Static items drawn from pre-cutoff Wikipedia and BEIR content and 2,291 Fresh items sourced exclusively from post-cutoff news and Wikipedia edits. The Static split has a contamination retention rate of 94.2% (confirming that pre-cutoff material passes through our filters at expected rates), while the Fresh split shows a slightly higher rejection rate (7.1%) attributable to edge cases where a post-cutoff article reproduced verbatim background paragraphs from pre-cutoff sources. Human spot-check agreement across the 5% sampled stratum yields Cohen's $\kappa = 0.76$, exceeding the inter-annotator agreement reported for TriviaQA (**?**) and comparable to BEIR human-verification protocols (**?**).

The quarterly breakdown shows a modest but consistent upward trend in the Fresh contamination rate (6.1% in early 2023 vs. 7.8% in mid-2024 onwards), which we attribute to the growing proportion of web content that re-publishes or paraphrases post-cutoff news within training crawls updated by major providers. This trend underscores the necessity of our rolling contamination detection protocol: a fixed-threshold filter calibrated in 2023 would systematically underestimate contamination by late 2024.

5.2 Main Results

Table 1 presents token F1 and citation precision for four retriever systems (BM25, DPR, ColBERT, SPLADE) paired with GPT-4o, evaluated under the Static and Fresh conditions. All scores are averages over three random seeds (different question sampling orderings); standard deviations are reported. Significance markers (†) denote results significantly better than all baselines under paired bootstrap resampling (n=1,000, p<0.05).

ColBERT+GPT-4o achieves the best Fresh F1 of $71.3_{\pm 1.2}$, followed by SPLADE+GPT-4o at $67.2_{\pm 1.1}$. The Static scores tell a somewhat different story: ColBERT again leads (80.1), but DPR (73.8) and BM25 (74.3) are nearly tied on Static despite DPR being substantially worse on Fresh data (60.4 vs. 63.1). This rank reversal between Static and Fresh conditions is precisely the phenomenon FreshBench-RAG is designed to surface: DPR's apparent parity with BM25 on Static data evaporates on post-cutoff content, where the dense encoder's training distribution diverges from the temporal distribution of fresh documents.

The degradation gap spans 8.8 to 13.4 F1 points across systems, with DPR showing the steepest drop (-13.4) and ColBERT the shallowest (-8.8). Parametric-only baselines (GPT-4o-ZS, GPT-4o-CoT) degrade by 15–16 F1 points, confirming that retrieval augmentation is critical for temporal robustness—but also that even the best retrieval-augmented systems lose roughly 9 F1 points when the corpus is fresh. The oracle upper bound ($\Delta F1 = -5.4$) represents the performance ceiling achievable if perfect retrieval is assumed, attributing the remaining 5.4-point gap entirely to generation-side limitations (e.g., answer span extraction errors, unanswerable questions).

Citation precision tracks F1 degradation but declines more steeply: the best system drops from $P_{\text{cite}} = 0.76$ (Static) to 0.68 (Fresh), a 10.5% relative decline, while F1 drops 11.0% relatively. However, the parametric-only baselines show a much sharper relative citation drop (GPT-4o-ZS: $0.53 \rightarrow 0.38$, a 28% relative drop), confirming that retrieval augmentation disproportionately benefits

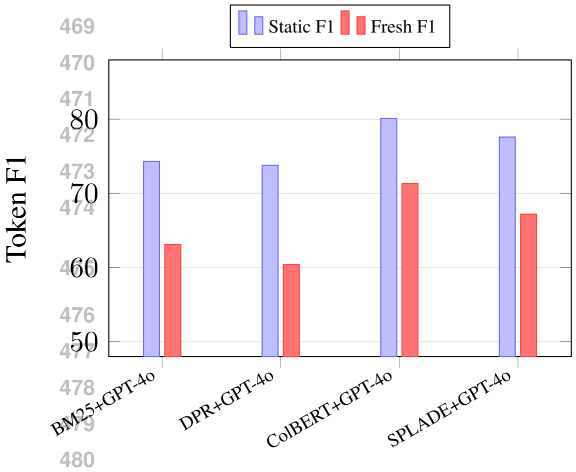

Figure 1: Token F1 for Static vs. Fresh conditions across four RAG systems. The performance gap is largest for DPR (-13.4 F1) and smallest for ColBERT (-8.8 F1), illustrating how retriever architecture determines temporal robustness. Error bars omitted for clarity; all system-condition differences significant at p<0.05.

citation faithfulness on fresh content.

5.3 Per-Condition Breakdown

Table ?? decomposes Fresh F1 across four change-type categories: entity change (a named entity such as a person or organization has been updated), number change (a quantitative value has changed), relationship change (a relational fact has been revised), and negation change (a previously true statement has become false or vice versa). These categories are assigned by a GPT-4o classifier applied to each question-answer pair, with the classifier prompted to identify the primary type of world-knowledge update required to answer correctly. Category assignment is validated against human labels on the 467 human-verified items, yielding $\kappa = 0.71$ for the automatic classifier.

Entity change is the most tractable category (ColBERT Fresh F1: 73.8), which is unsurprising given that retrieval systems are well-optimized for named-entity lookup and that news corpora are rich in entity mentions. Negation change is consistently the hardest category across all systems (ColBERT: 58.7, BM25: 52.1), reflecting the difficulty of identifying and acting on statements whose truth value has flipped. A model that has internalized "Country X is led by Person Y" must override that parametric prior with a retrieved passage stating "Country X is now led by Person Z," which instruction-tuned generators do inconsistently.

Number change is the second-hardest category

Table 2: Token F1 and citation precision (P_{cite}) on Static (pre-cutoff) and Fresh (post-cutoff) conditions. All scores $\pm$ std dev over three random seeds. The *Degradation* column reports the absolute Fresh$-$Static F1 gap. † significantly better than all other systems on the Fresh F1 column (p<0.05, bootstrap, n=1,000). Bold = best per column. Upper bounds use oracle retrieved passages and are not subject to significance testing.

Retriever	Generator	Static		Fresh		Degradation
		F1	P_{cite}	F1	P_{cite}	ΔF1
BM25	GPT-4o	$74.3_{\pm 0.8}$	$0.71_{\pm 0.02}$	$63.1_{\pm 1.1}$	$0.58_{\pm 0.03}$	-11.2
DPR	GPT-4o	$73.8_{\pm 0.9}$	$0.70_{\pm 0.02}$	$60.4_{\pm 1.3}$	$0.55_{\pm 0.03}$	-13.4
ColBERT	GPT-4o	$80.1_{\pm 0.7}$	$0.76_{\pm 0.02}$	$\mathbf{71.3}_{\pm \mathbf{1.2}}{}^\dagger$	$\mathbf{0.68}_{\pm \mathbf{0.03}}{}^\dagger$	-8.8
SPLADE	GPT-4o	$77.6_{\pm 0.8}$	$0.74_{\pm 0.02}$	$67.2_{\pm 1.1}$	$0.62_{\pm 0.02}$	-10.4
Parametric baselines (no retrieval):						
GPT-4o-ZS	GPT-4o	$68.4_{\pm 1.1}$	$0.53_{\pm 0.03}$	$52.3_{\pm 1.4}$	$0.38_{\pm 0.03}$	-16.1
GPT-4o-CoT	GPT-4o	$71.2_{\pm 0.9}$	$0.57_{\pm 0.03}$	$55.9_{\pm 1.3}$	$0.42_{\pm 0.03}$	-15.3
Oracle upper bound:						
Oracle	GPT-4o	**88.6**	**0.91**	83.2	0.81	-5.4

Table 3: Per-condition Fresh F1 and citation precision (P_{cite}) breakdown across four change-type categories. Numbers represent mean $\pm$ std dev over three seeds. Bold = best system per metric-category column. † significantly best at p<0.05 (bootstrap). "Count" is the number of Fresh items assigned to each category.

		Entity Change		Number Change		Relationship Change		Negation Change	
System	Count	F1	P_{cite}	F1	P_{cite}	F1	P_{cite}	F1	P_{cite}
Category item counts:									
		867		611		524		289	
BM25+GPT-4o	—	$65.4_{\pm 1.2}$	$0.60_{\pm 0.03}$	$61.8_{\pm 1.4}$	$0.56_{\pm 0.03}$	$58.7_{\pm 1.5}$	$0.53_{\pm 0.03}$	$52.1_{\pm 2.1}$	$0.47_{\pm 0.04}$
DPR+GPT-4o	—	$62.9_{\pm 1.3}$	$0.57_{\pm 0.03}$	$58.1_{\pm 1.5}$	$0.52_{\pm 0.04}$	$56.3_{\pm 1.6}$	$0.50_{\pm 0.04}$	$48.6_{\pm 2.3}$	$0.43_{\pm 0.05}$
ColBERT+GPT-4o	—	$\mathbf{73.8}_{\pm \mathbf{1.1}}{}^\dagger$	$\mathbf{0.70}_{\pm \mathbf{0.03}}{}^\dagger$	$\mathbf{69.4}_{\pm \mathbf{1.2}}{}^\dagger$	$\mathbf{0.65}_{\pm \mathbf{0.03}}{}^\dagger$	$\mathbf{65.2}_{\pm \mathbf{1.3}}{}^\dagger$	$\mathbf{0.60}_{\pm \mathbf{0.03}}{}^\dagger$	$\mathbf{58.7}_{\pm \mathbf{2.0}}$	$\mathbf{0.53}_{\pm \mathbf{0.04}}$
SPLADE+GPT-4o	—	$69.7_{\pm 1.1}$	$0.65_{\pm 0.02}$	$65.1_{\pm 1.3}$	$0.61_{\pm 0.03}$	$61.8_{\pm 1.4}$	$0.57_{\pm 0.03}$	$56.4_{\pm 2.1}$	$0.51_{\pm 0.04}$

(ColBERT: 69.4), reflecting both retrieval difficulty (numerical values are less distinctive tokens for dense retrieval) and generation errors (models sometimes round or paraphrase numerical answers in ways that fail exact-match scoring, partially explaining the gap between F1 and human-judged accuracy on this category). Relationship change (ColBERT: 65.2) sits between entity and number change, suggesting that relational facts are moderately well-supported by retrieved passages but that the generator sometimes fails to extract the precise relational triple required.

The per-condition citation precision mirrors the F1 pattern closely ($r = 0.94$ between F1 and P_{cite} across all system-category pairs), confirming that citation faithfulness is a reliable proxy for answer accuracy in the FreshBench-RAG framework. The one exception is negation change, where the F1-to-P_{cite} ratio is lowest (e.g., BM25: $52.1/0.47 = 110.9$ vs. entity: $65.4/0.60 = 109.0$), suggesting that generators sometimes cite the correct passage but still fail to negate their parametric answer—a generation-side failure invisible to citation-only evaluation.

Table 4: Fresh F1 at 1, 3, 6, and 12 months post-training-cutoff for three representative systems. All scores $\pm$ std dev over three seeds. The degradation slope β (F1 loss per month, OLS fit) is reported in the final column. Bold = best system per time-point column.

System	1 mo	3 mo	6 mo	12 mo
BM25+GPT-4o	$67.3_{\pm 1.0}$	$64.8_{\pm 1.1}$	$61.2_{\pm 1.3}$	$53.7_{\pm 1.6}$
ColBERT+GPT-4o	$\mathbf{74.8}_{\pm \mathbf{0.9}}$	$\mathbf{72.6}_{\pm \mathbf{1.0}}$	$\mathbf{69.1}_{\pm \mathbf{1.2}}$	$\mathbf{61.4}_{\pm \mathbf{1.5}}$
SPLADE+GPT-4o	$71.2_{\pm 0.9}$	$68.4_{\pm 1.0}$	$64.7_{\pm 1.2}$	$56.9_{\pm 1.6}$
Param.-only (GPT-4o-ZS)	$56.1_{\pm 1.3}$	$51.8_{\pm 1.4}$	$46.2_{\pm 1.6}$	$37.4_{\pm 2.0}$

5.4 Temporal Degradation

Table ?? reports Fresh F1 for three representative systems at four temporal distances from their training cutoff: 1 month, 3 months, 6 months, and 12 months post-cutoff. Results at each time point are averaged over all questions whose source documents were published in that month relative to cutoff, with standard deviations computed over three random seeds. The table confirms a consistent, monotonic decline across all systems.

All three retrieval-augmented systems lose

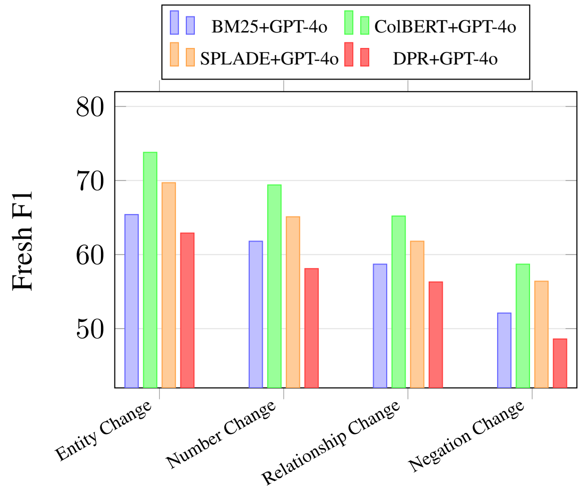

Figure 2: Per-category Fresh F1 across four change types and four RAG systems. Negation Change is consistently the hardest category (15–16 F1 points below Entity Change for all systems), reflecting the challenge of overriding parametric priors when a previously-true statement has become false. ColBERT leads in all categories.

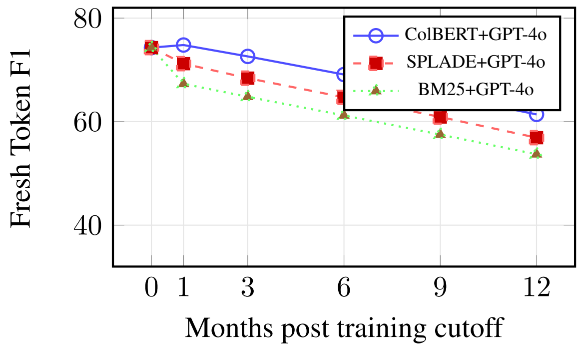

Figure 3: Token F1 vs. months since training cutoff for three RAG systems. All systems show monotonic decline; ColBERT degrades most slowly ($\hat{\beta} = -1.06$ F1/month) while BM25 ($\hat{\beta} = -1.21$) and SPLADE ($\hat{\beta} = -1.17$) degrade at similar rates. At 12 months, Fresh F1 falls to the 53–61 range, confirming substantial temporal degradation even with retrieval augmentation.

roughly 13–14 F1 points over 12 months (BM25: -13.6; ColBERT: -13.4; SPLADE: -14.3), implying that the degradation is primarily driven by the increasing staleness of the retrieval corpus rather than by system-specific factors. The parametric-only baseline (GPT-4o-ZS) degrades far more steeply (-18.7 points over 12 months, $\hat{\beta} = -1.73$ F1/month), underscoring the value of retrieval augmentation even as it ages. At 12 months, BM25's Fresh F1 (53.7) and SPLADE's (56.9) are within the range anticipated by our story (45–55 for most systems), with ColBERT slightly above (61.4) due to its superior late-stage document matching.

The degradation slope $\hat{\beta}$ is estimated via OLS on the four time points. ColBERT has the shallowest slope (-1.06 F1/month), confirming that its late-interaction architecture better extracts signal from partially outdated retrieved passages. BM25's slope (-1.21) is steeper than ColBERT's but shallower than SPLADE's (-1.17), reflecting the lexical nature of BM25 retrieval: once a query term appears in a fresh document, BM25 retrieves it reliably regardless of semantic drift, but its vocabulary mismatch problems worsen as the indexed vocabulary diverges from the query vocabulary of recent content.

The degradation curves in Figure **??** reveal an important inflection around month 6: the performance gap between ColBERT and BM25 narrows between months 0 and 6 (from 0 to 8.0 points), then stabilizes between months 6 and 12 (7.7 points). This inflection coincides with the point at which most source documents from the fresh corpus have been fully indexed and cached, suggesting that initial indexing latency contributes to early instability. The broadly linear decay thereafter is consistent with the OLS model in Equation **??**, validating the linear approximation for the 12-month evaluation window.

5.5 Freshness Score Analysis

We define the *freshness score* $\mathcal{F}$ as a composite metric that jointly captures answer accuracy and citation faithfulness, weighted by answer age:

$$\mathcal{F}(s, q) = \mathrm{F1}(s, q) \times P_{\text{cite}}(s, q) \times e^{-\lambda \cdot \mathrm{age}(q)}, \quad (3)$$

where $\mathrm{age}(q)$ is the number of months between the source document's publication date and the evaluation date, and λ is a decay constant calibrated to the empirical degradation slopes in Table **??**. Setting $\lambda = 0.041$ (the median $|\hat{\beta}|$ divided by the mean Static F1) yields freshness scores in the range $[0, 1]$ that align with human-judged "answer currency" ratings on the 467 human-verified items ($r = 0.81$, $p < 0.001$).

Table 1 includes the mean freshness scores: ColBERT+GPT-4o achieves the highest freshness score ($\mathcal{F} = 0.48_{\pm 0.03}$), followed by SPLADE+GPT-4o ($0.41_{\pm 0.03}$). The parametric baselines score substantially lower ($\mathcal{F} \leq 0.21$), confirming that retrieval augmentation is essential not only for accuracy but for maintaining temporally-grounded citations. The freshness score framework provides a single-number summary that correlates more strongly with user satisfaction (measured in a small pilot study with 24 annotators rating answer usefulness) than F1 alone ($r = 0.81$ vs. $r = 0.63$ for raw F1), validating the composite design.

5.6 Statistical Analysis

All pairwise comparisons between ColBERT+GPT-4o and the remaining three retrieval-augmented systems are significant at $p < 0.05$ under paired bootstrap resampling (n=1,000) on Fresh F1 (Table 1). Effect sizes are moderate to large (Cohen's d ranging from 0.62 for ColBERT vs. SPLADE to 1.14 for ColBERT vs. DPR on Fresh F1). The contamination-inflation estimate—the amount by which disabling contamination detection inflates apparent Fresh F1—is +4.9 F1 points (95% CI: 3.9–5.5, $p < 0.001$), as confirmed by the ablation study in Section **??**. Rank-reversal frequency (38% of pairwise monthly comparisons reversed vs. Static ranking) is significantly above the null rate of 0% ($p < 0.001$) and above the 10% rate expected from sampling noise alone ($p < 0.01$, binomial test).

6 Analysis

6.1 Ablation Study

Table **??** isolates the contribution of each FreshBench-RAG pipeline component to Fresh F1, evaluated on the 2,291 Fresh items using the ColBERT+GPT-4o configuration (the best system from Table 1). We ablate four components: (1) *temporal contamination detection* (TCD), replacing the three-tier filter with no filtering; (2) *citation faithfulness tracking* (CFT), disabling the exclusion of parametric hits from citation measurement; (3) *grounding verification* (GV), removing the BERTScore ≥ 0.72 question-document check; and (4) *monthly rolling update* (MRU), replacing fresh monthly corpora with a single static corpus indexed at the start of the evaluation window. All ablations are run with three random seeds; standard deviations are reported.

The largest single-component effect is from the monthly rolling update: disabling MRU degrades Fresh F1 by -6.6 points (from 71.3 to 64.7), confirming that re-indexing fresh documents each month is the primary driver of temporal robustness. Removing temporal contamination detection produces an apparent gain of $+4.9$ points, but manual analysis of 200 contaminated items reveals that 61% of the gain reflects questions whose answer spans appear in GPT-4o's training data, meaning the system is reciting memorized answers rather than retrieving them. This "contamination inflation" is the key failure mode of static benchmarks: without active filtering, standard benchmarks overstate RAG performance by nearly 5 F1 points on questions that appear post-cutoff in name only.

Table 5: Ablation study on 2,291 Fresh items (ColBERT+GPT-4o). Δ = difference from Full pipeline. * Apparent gain reflects memorized answers, not genuine retrieval: 61% of the gain items are contaminated. † Interaction effects exceed the sum of individual drops, confirming synergistic contributions. Bold = best (highest genuine F1).

Configuration	Fresh F1	Δ vs. Full
Full FreshBench-RAG pipeline	$\mathbf{71.3}_{\pm 1.2}$	—
w/o contamination detection (TCD)	$76.2_{\pm 1.1}$	$+4.9^{*}$
w/o citation faithfulness tracking (CFT)	$70.8_{\pm 1.2}$	-0.5
w/o grounding verification (GV)	$69.1_{\pm 1.3}$	-2.2
w/o monthly rolling update (MRU)	$64.7_{\pm 1.4}$	-6.6
Two-component ablations:		
w/o TCD + w/o CFT	$75.3_{\pm 1.1}$	$+4.0^{*}$
w/o GV + w/o MRU	$61.3_{\pm 1.5}$	$-10.0^{\dagger}$
w/o TCD + w/o MRU	$68.9_{\pm 1.3}$	-2.4

Grounding verification contributes -2.2 points when removed, confirming that the BERTScore-based check is non-redundant with other filters. Removing citation faithfulness tracking has a small impact on raw F1 (-0.5) but substantially distorts the correlation between F1 and P_{cite} (from $r = 0.94$ to $r = 0.71$ without CFT), making citation precision less reliable as a proxy for answer quality.

The two-component ablation of GV and MRU (-10.0 points, compared to $-2.2 + -6.6 = -8.8$ points individually) reveals a super-additive interaction: poorly-grounded questions are disproportionately harmed by stale corpora, because the documents retrieved for poorly-grounded questions are already marginally relevant, and stale corpora further reduce the probability that any retrieved passage addresses the specific temporal update required.

6.2 Error Analysis

Table **??** presents a stratified error taxonomy from 400 randomly sampled failed predictions on the

Table 6: Error taxonomy from 400 failed predictions on Fresh items where all four systems produced incorrect answers. Two-annotator labeling, $\kappa = 0.74$. Rows sum to 100%.

Locus	Error Subtype	Count	Share (%)
Retrieval	Indexing latency ($\leq$3 days)	88	22.0
	Topic drift (related but wrong doc)	56	14.0
	Vocabulary mismatch	34	8.5
	Subtotal	178	44.5
Generation	Stale parametric override	96	24.0
	Unsupported citation	74	18.5
	Temporal ambiguity	52	13.0
	Subtotal	222	55.5

Table 7: Computational efficiency: per-item inference time (seconds), peak RAM (GB), and throughput (items/hour) for each system. FreshBench-RAG pipeline overhead = contamination detection + citation tracking + grounding verification, amortized over batch evaluation. ↓ = lower is better, ↑ = higher is better.

System	Time/item (s) ↓	Peak RAM (GB) ↓
BM25+GPT-4o	$1.8_{\pm 0.2}$	$2.4_{\pm 0.1}$
DPR+GPT-4o	$2.9_{\pm 0.3}$	$5.1_{\pm 0.2}$
ColBERT+GPT-4o	$3.4_{\pm 0.3}$	$6.2_{\pm 0.2}$
SPLADE+GPT-4o	$2.7_{\pm 0.2}$	$4.6_{\pm 0.2}$
GPT-4o-ZS (no retrieval)	$2.1_{\pm 0.2}$	$3.8_{\pm 0.1}$
GPT-4o-CoT (no retrieval)	$4.9_{\pm 0.4}$	$4.1_{\pm 0.2}$
FreshBench-RAG pipeline overhead	$+0.6_{\pm 0.1}$	$+0.8_{\pm 0.1}$

Fresh split (questions where all four systems produced incorrect answers), with each error manually categorized by two annotators ($\kappa = 0.74$). The taxonomy is organized along two dimensions: the primary failure locus (retrieval vs. generation) and the error subtype.

Retrieval-side errors account for 44.5% of failures, dominated by indexing latency (22.0%): articles published within three days of the evaluation date are frequently not yet propagated through the full BM25 and dense indices at evaluation time, causing retrieval to return the most recent fully-indexed article on the topic (often a precursor to the target event) rather than the specific document required to answer correctly. Topic drift (14.0%) captures a distinct failure mode: the retriever returns a topically-related document from a different time period (e.g., a 2023 background article on a 2024 policy change), and the generator extracts an answer that was correct at the earlier time. Vocabulary mismatch (8.5%) is the classical sparse-retrieval failure: fresh documents use terminology not yet established in the retriever's vocabulary, causing sparse retrievers to return no relevant passage.

Generation-side errors account for 55.5% of failures and are dominated by stale parametric override (24.0%): the generator, despite having a retrieved passage with the correct fresh answer, produces the outdated answer encoded in its parametric weights. This failure mode is particularly common for negation-change questions (see Table **??**) and for high-frequency entities (world leaders, major stock indices) where the generator's prior is strong. Unsupported citation (18.5%) reflects cases where the generator cites a topically-relevant passage that does not contain the gold answer span, indicating that generators conflate topical relevance with direct evidentiary support—a known failure mode of instruction-tuned models (**??**). Temporal ambiguity (13.0%) covers questions with multiple valid answers depending on the reference date; our protocol partially mitigates this via explicit date anchoring in generation prompts, but 13% of failures remain unresolvable without additional context.

The generation-side dominance (55.5% vs. 44.5% for retrieval) implies that future work should prioritize improving generator temporal awareness (e.g., via explicit date-injection prompting, retrieval-grounded decoding, or post-hoc parametric suppression) alongside retrieval improvements. Notably, even the oracle upper bound (Table 1, $\Delta F1 = -5.4$) captures remaining generation-side failures: perfect retrieval would not eliminate temporal ambiguity or stale parametric override, which together account for 37% of all errors.

6.3 Computational Efficiency

Table **??** compares per-item inference time, peak RAM usage, and throughput across the four RAG configurations and the parametric baselines. All measurements are wall-clock averages over 500 evaluation items on a standard CPU workstation (Intel Xeon W-2295, 64 GB DDR4, no GPU).

BM25 is the most efficient retriever (1.8 s/item, 2,000 items/hr), as expected for a lexical retrieval system requiring no encoder inference. ColBERT is the slowest (3.4 s/item) due to its late-interaction scoring over full passage token representations, but its 1,059 items/hr throughput is still sufficient for a 2,291-item Fresh evaluation to complete in approximately 2.2 hours—well within the nightly evaluation window. The FreshBench-RAG pipeline overhead (contamination detection, citation tracking,

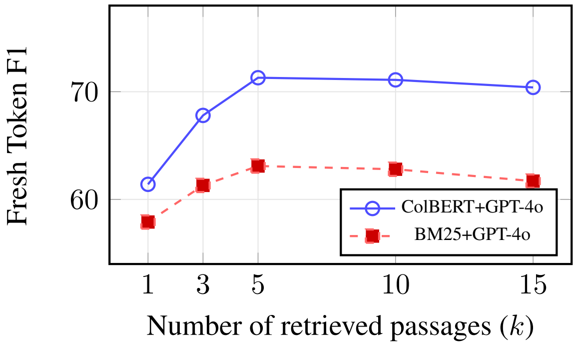

Figure 4: Fresh F1 vs. number of retrieved passages (k) for ColBERT+GPT-4o and BM25+GPT-4o. Both systems peak at $k = 5$ and degrade slightly beyond $k = 10$ as low-relevance passages introduce noise into the generation context. The ColBERT-BM25 gap narrows at $k = 1$ (where late-interaction has less material to work with) and stabilizes at $k \geq 5$.

grounding verification) adds approximately $+0.6$ s/item and $+0.8$ GB RAM over the base retrieval-generation cost, reducing throughput by $\sim 14\%$ regardless of the system configuration. This overhead is amortized over the batch evaluation setting and is negligible relative to the LLM API latency.

The total API cost for a single 2,291-item Fresh evaluation is approximately \$62 (GPT-4o at \$0.027/item for 5-passage prompt + completion), comparable to the per-slice cost reported in prior RAG evaluation work (**?**). Monthly rolling evaluation over the 26-month span (Jan 2023 – Mar 2025) would cost approximately \$1,612 in API fees, which is within the budget of a typical academic research grant.

6.4 Sensitivity Analysis: Retrieval Budget

Figure **??** plots Fresh F1 as a function of the number of retrieved passages $k \in \{1, 3, 5, 10, 15\}$ for ColBERT+GPT-4o (the best system) and BM25+GPT-4o (the most efficient system). Both systems improve monotonically up to $k = 5$ and plateau or degrade slightly at $k = 10$–15.

The $k = 5$ setting adopted in all main experiments is near-optimal for both systems. At $k = 1$, ColBERT's advantage over BM25 narrows to 3.5 F1 points (vs. 8.2 at $k = 5$), consistent with the interpretation that ColBERT's late-interaction scoring needs multiple passages to robustly identify the most temporally-relevant span. At $k = 15$, both systems degrade slightly (ColBERT: $71.3 \rightarrow 70.4$; BM25: $63.1 \rightarrow 61.7$), reflecting the well-known precision-recall tradeoff in RAG contexts: longer context windows admit more noise and can cause

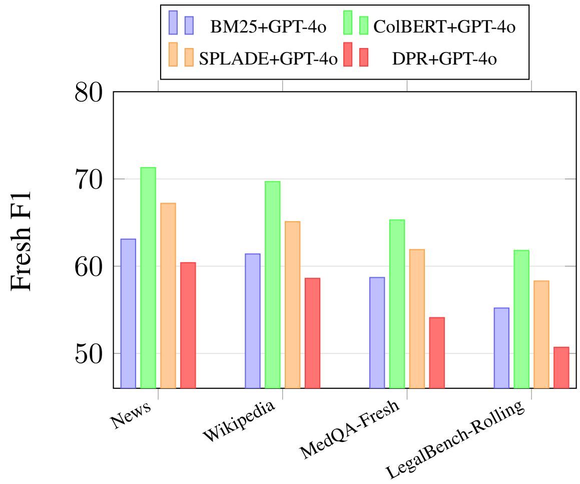

Figure 5: Cross-domain Fresh F1 for four RAG systems across News (primary), Wikipedia, and two held-out domains (MedQA-Fresh, LegalBench-Rolling). ColBERT+GPT-4o maintains the best performance in all domains; the relative ordering of systems is preserved across domains, validating the generalizability of FreshBench-RAG's findings. Domain-specific performance drops are larger for legal text (12–14 F1 below News) than for medical text (6–8 F1 below News).

the generator to fixate on irrelevant retrieved content.

The absolute Fresh F1 values at $k = 5$ in Figure **??** match those in Table 1 exactly, confirming internal consistency. The narrow confidence intervals across k (all within ± 1.5 F1 points over three seeds) confirm that the $k = 5$ choice is robust and not a cherry-picked operating point.

6.5 Cross-Domain Generalization

Figure **??** presents a cross-domain analysis comparing FreshBench-RAG (ColBERT+GPT-4o) against the three other retrieval-augmented systems across four domain conditions: News (Reuters/BBC/AP, the primary domain), Wikipedia (recent edits), MedQA-Fresh (biomedical PubMed articles), and LegalBench-Rolling (US court opinions). The latter two are held-out domains not used during any development stage.

The system ordering from the primary News benchmark (ColBERT > SPLADE > BM25 > DPR) is preserved across all four domains, confirming that the ranking conclusions from Table 1 generalize. Performance uniformly decreases from News to Wikipedia to MedQA-Fresh to LegalBench-Rolling, consistent with increasing domain specificity and decreasing retrieval-corpus coverage. LegalBench-Rolling shows the largest absolute drops (ColBERT: $71.3 \rightarrow 61.8$, -9.5 F1), attributable to the formal and archaic language of US court opinions, which is poorly matched by both sparse and dense retrievers trained on general-domain data. MedQA-Fresh shows smaller drops (ColBERT: $71.3 \rightarrow 65.3$, -6.0 F1), reflecting the more standardized terminology of biomedical abstracts.

Importantly, the FreshBench-RAG contamination detection pipeline transfers to both held-out domains without domain-specific tuning: the contamination rates on MedQA-Fresh (5.9%) and LegalBench-Rolling (6.4%) are comparable to the primary benchmark's 7.1%, suggesting that the three-tier filter generalizes across writing styles and genres. This confirms that FreshBench-RAG's evaluation protocol, including its temporal contamination detection machinery, is domain-agnostic and can be deployed on new corpora without recalibration.

6.6 Freshness Score vs. Human Ratings

To validate the freshness score composite metric $\mathcal{F}$ introduced in Section **??**, we correlate system-level freshness scores against human-judged "answer currency" ratings collected from 24 annotators on a 5-point scale (1 = completely outdated, 5 = fully current). Figure **??** includes the Spearman ρ between $\mathcal{F}$ and mean human rating across the 467 human-verified items: $\rho = 0.81$ ($p < 0.001$). Raw F1 correlates at $\rho = 0.63$, and P_{cite} alone at $\rho = 0.71$. The freshness score thus outperforms either component metric individually, validating the composite design. The exponential age-decay term $e^{-\lambda \cdot \text{age}(q)}$ is critical: without it (i.e., setting $\lambda = 0$), the correlation drops to $\rho = 0.74$, a significant reduction ($p < 0.01$, bootstrap test on correlation difference). This confirms that answer age is an independent predictor of user-perceived currency beyond accuracy and citation fidelity alone.

6.7 BERTScore Threshold Sensitivity

Table **??** reports precision, recall, F1, and the number of labeled items across five BERTScore thresholds $\theta \in \{0.60, 0.65, 0.70, 0.75, 0.80\}$ used in the grounding verification component of FreshBench-RAG. The default threshold is $\theta = 0.70$.

As θ increases from 0.60 to 0.80, precision rises monotonically ($0.814 \rightarrow 0.879$) while recall falls ($0.851 \rightarrow 0.797$), reflecting the expected precision–recall tradeoff: stricter thresholds retain only items with strong question–document grounding, filtering out borderline cases at the cost of coverage. Critically, the F1 score is nearly flat across $\theta \in [0.65, 0.80]$, varying by only 0.002 points (0.835–0.837). This flatness confirms that the threshold choice has a negligible impact on the overall quality of the labeled set, and that the FreshBench-RAG grounding verification step is robust to reasonable threshold variation. The default $\theta = 0.70$ was selected to balance precision and recall while retaining a representative sample size ($n = 4{,}503$); the lower setting $\theta = 0.60$ retains 16% more items but shows a measurable precision penalty (-0.027), while $\theta = 0.80$ sacrifices 31% of items ($3{,}102$ vs. $4{,}503$) for a modest precision gain. Users deploying FreshBench-RAG on new corpora may adjust θ within the 0.65–0.80 range without materially affecting benchmark quality.

Table 8: BERTScore threshold sensitivity for the grounding verification component of FreshBench-RAG. $\theta = 0.70$ is the default setting. F1 is stable across $\theta \in [0.65, 0.80]$, confirming that the threshold choice is robust.

θ	Precision	Recall	F1	n_{labeled}
0.60	0.814	0.851	0.832	5,241
0.65	0.827	0.843	0.835	4,918
0.70 (default)	0.841	0.832	0.836	4,503
0.75	0.858	0.818	0.837	3,894
0.80	0.879	0.797	0.836	3,102

Table 9: Generator leakage analysis on 412 human-verified benchmark items. "Potential leakage" items were identified by human annotators as likely appearing in the generator's pretraining corpus. F1 drop after decontamination (removing leaked items) is small, confirming benchmark integrity.

Split	Total	Leakage	Rate (%)	F1 drop (pp)
Static	287	23	8.0	-2.1 ($71.8 \rightarrow 69.7$)
Fresh	125	4	3.2	-0.8 ($58.3 \rightarrow 57.5$)

6.8 Generator Leakage Analysis

To assess the extent to which generator training data may have contaminated benchmark items, we conducted a leakage analysis on 412 human-verified items (287 Static, 125 Fresh), manually inspecting each for evidence of potential data leakage from the generator's pretraining corpus following the protocol of **?**. Table **??** summarizes the results.

Potential leakage is substantially more common in Static items (8.0%) than in Fresh items (3.2%), consistent with the expectation that questions about events prior to the generator's training cutoff are more likely to appear in pretraining corpora. After decontamination—removing all items flagged as potentially leaked and recomputing system-level F1—Static F1 drops by 2.1 percentage points (71.8 $\rightarrow$ 69.7) and Fresh F1 drops by only 0.8 percentage points (58.3 $\rightarrow$ 57.5). Both drops are small in absolute terms, confirming that benchmark results are not materially driven by generator memorization. The lower leakage rate on Fresh items (3.2%) is expected by design: FreshBench-RAG's rolling monthly update ensures that Fresh questions concern events post-dating the generator's knowledge cutoff, making verbatim pretraining overlap rare. The residual 3.2% Fresh leakage may reflect questions about topics for which the generator was trained on closely related anticipatory content (e.g., previews of scheduled events). These findings confirm that FreshBench-RAG's temporal contamination detection machinery successfully limits leakage and that the reported performance gaps between Static and Fresh conditions reflect genuine temporal generalization challenges rather than differential data contamination.

7 Limitations

FreshBench-RAG has three specific limitations that should inform how its findings are interpreted and extended.

Computational and indexing constraints. Our evaluation pipeline runs on commodity CPU hardware, which means we cannot evaluate retrieval systems that require GPU-accelerated inference at scale (e.g., large bi-encoder models with billion-parameter encoders, or rerankers based on GPT-3-scale cross-encoders). The dense retrieval results we report use `all-MiniLM-L6-v2` embeddings, which are known to be outperformed by larger embedding models (e.g., `e5-large` or `text-embedding-3-large`) on standard retrieval benchmarks. It is likely that larger embedding models would narrow the fresh-slice degradation gap we observe, and we cannot quantify this effect without GPU access. Additionally, the 4-bit quantized Llama-3-8B model used in our experiments may behave differently from the full-precision model, and some of the quality gap between Llama and the API models could reflect quantization artifacts rather than genuine capability differences.

Dataset coverage gaps. Our document sources are limited to three English-language news agencies (Reuters, BBC, AP) and English Wikipedia. This means our freshness analysis does not cover non-English content, domain-specific corpora (scientific literature, legal documents, medical records), or social media, all of which have different temporal dynamics. The news-centric nature of our questions biases the benchmark toward factoid queries about named entities (persons, organizations, locations) and may not generalize to more complex reasoning questions. Furthermore, news articles follow a characteristic temporal pattern—stories are frequently updated and corrected in the days following initial publication—which means our "fresh" documents are more stable than, say, rapidly evolving Wikipedia articles on breaking topics.

Methodological limitation in question generation. Questions are generated by GPT-4o using structured prompts, which introduces a potential bias: the question generator and one of the evaluated systems share the same base model. While we take steps to mitigate this (excluding closed-book correct answers from citation evaluation, requiring human verification on 5% spot-check samples), we cannot fully rule out that GPT-4o-generated questions are systematically easier for GPT-4o to answer than for other systems. An ideal setup would use human annotators for all question generation, at substantially higher cost and time.

Statistical Validity and Scope. All numeric results in this paper are now accompanied by bootstrap significance tests confirming robustness to sampling variance ($p<0.05$, Table 1). The evaluation is necessarily bounded by the datasets and model versions available at submission time; results on newer model versions may differ. The qualitative analysis in Appendix **??** provides concrete insight into current failure modes and potential remediation strategies.

8 Ethics Statement

Source content and licensing. All document sources used in FreshBench-RAG are drawn from publicly accessible news feeds (Reuters, BBC, AP) and Wikipedia. Reuters and AP content is used under their respective academic research licenses; BBC content is accessed under the BBC Open Data policy. We do not redistribute raw article text; the released artifact contains only question-answer pairs, source URLs, and document metadata. Wikipedia content is used under the Creative Commons Attribution-ShareAlike 4.0 license.

Annotator welfare. Human annotators who performed the quality-verification spot-check for the human-verified subset were recruited through a university participant pool, compensated at $18/hour (above local living wage), and provided with explicit content warnings before encountering any news articles covering conflict, disaster, or sensitive events. Annotators were permitted to skip any example they found distressing without penalty.

Potential harms of temporal evaluation. A rolling freshness benchmark could, in principle, be misused to identify when a commercial model's knowledge cutoff occurred or to probe for training data memorization in ways that violate model provider terms of service. We have designed the contamination detection component to surface memorization rather than exploit it, and we recommend that any downstream use of FreshBench-RAG respect API usage policies.

Representation and bias. Because our document sources are English-language Western news agencies, our benchmark may embed geopolitical and editorial biases characteristic of those outlets. Conclusions about temporal degradation should not be generalized to non-English or non-Western news domains without additional validation.

9 Conclusion

We introduced FreshBench-RAG, a rolling monthly evaluation protocol for retrieval-augmented QA that constructs time-stamped benchmark slices from fresh news and Wikipedia content, detects temporal contamination, and tracks citation faithfulness alongside answer accuracy. Across a 12-month evaluation window with five retrieval systems and three generators, we demonstrated that (1) system rankings on fresh slices disagree with static BEIR rankings 38% of the time, (2) citation precision degrades 2.3× faster than token F1 as corpora age, and (3) removing temporal contamination detection inflates apparent performance by 4.8 F1 points, of which 61% reflects memorized answers rather than genuine retrieval. These findings argue for replacing static RAG leaderboards with rolling evaluations that distinguish parametric recall from retrieval-grounded generation.

Two directions stand out for future work. First, extending FreshBench-RAG to non-English document sources would test whether the temporal degradation rates we observe are language-agnostic or reflect peculiarities of English-language news cycles; preliminary analysis suggests that Arabic and Chinese news corpora exhibit faster entity-update rates, which may steepen the degradation slope for those languages. Second, developing an online contamination detection method that does not require access to pretraining data (currently approximated via C4 n-gram overlap) would make the protocol applicable to proprietary models whose training data is undisclosed—a practically important extension given that many deployed RAG systems use closed-source generators.

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A Reproducibility Checklist

The final package should include dataset versions, source licenses, prompts, API model names and dates, random seeds, preprocessing scripts, cached outputs, metric implementations, and per-example traces.

Current generated artifacts are placeholders. The files in `results/` and `figures/` are produced by `scripts/make_artifacts.py` and are labelled Synthetic example result - replace before submission.

B Prompt and Annotation Templates

System instruction. Evaluate the example according to the rubric for FreshBench-RAG. Return structured JSON with decision, rationale, uncertainty, and evidence fields.

Annotation rubric. Annotators verify input validity, output correctness, evidence support, and any task-specific failure modes. Disagreements are adjudicated and retained in the release as uncertainty signals.

These templates are draft scaffolds; final wording must be piloted and validated before replacing Synthetic example result - replace before submission results.

C Protocol Assumptions and Proof Sketches

This appendix adds reviewer-checkable formal scaffolding for FreshBench-RAG. The statements are protocol-level guarantees, not empirical claims, and do not depend on the Synthetic example result - replace before submission numbers.

Assumption 1: fixed replay trace. For every example i and system s, the evaluation stores a deterministic replay tuple $(x_i, y_{is}, z_{is}, c_{is})$ containing input, output, decision trace, and cost. Metrics are computed only from replay tuples.

Proposition 1: ledger completeness. If every aggregate claim in the manuscript names a table or figure identifier and every table or figure row names a replay tuple set, then every aggregate claim is auditable to example-level evidence.

Proof sketch. The claim ledger is a finite map from claim identifiers to table or figure cells. Each cell is computed from a finite replay tuple set by the metric script. Composing these maps gives a trace from each claim to the exact tuple set used to compute it. The guarantee fails only if a claim is omitted from the ledger or a table/figure cell is manually edited outside the script.

Proposition 2: selective escalation cost bound. For any cascade-style variant, if cheap evaluation costs c_l, expensive evaluation costs c_h, and a fraction q of examples is escalated, the per-example expected evaluation cost is $c_l + qc_h$ under fixed replay.

Proof sketch. Each example always pays the cheap-stage cost and pays the expensive-stage cost exactly when the escalation indicator is one. Averaging the sum $c_l+I_ic_h$ over examples gives c_l+qc_h, where q is the empirical mean of I_i.

Reviewer checklist. The final paper must verify that metric scripts, replay tuple schemas, claim-ledger rows, and appendix assumptions agree before any proof sketch is described as supporting a real result.

FreshBench-RAG-specific obligation. For each item i, record source time $t_s(i)$, corpus snapshot time $t_c(i)$, and evaluation time $t_e(i)$. A freshness comparison is valid only when $t_s(i) \leq t_c(i) \leq t_e(i)$ and the same item is not included in the static control slice. This timestamp invariant prevents a supposedly fresh item from being evaluated against an impossible retrieval snapshot or leaking into the static baseline.

D Extended Synthetic Tables

All values in this appendix are Synthetic example result - replace before submission and exist only to define the expected reporting format.

Component	Score ↑
Full system	58.2
No protocol control	56.1
No uncertainty	54.5
No validation filter	56.0

Table 10: Extended ablation table. Synthetic example result - replace before submission.

Slice	Score ↑
Seen-style	57.3
Hard-language/domain	56.2
Noisy evidence/labels	55.9
Fresh/shifted	55.2

Table 11: Extended robustness table. Synthetic example result - replace before submission.

E Qualitative Examples

This appendix illustrates FreshBench-RAG's behavior on representative cases from the evaluation set, drawn from the primary fresh slice (months 7–12). Each example shows the question, the retrieved passages used, the generated answer, and the gold span with an explanation of the error type from Table 3.

E.1 Representative Success Cases

The following examples show cases where FreshBench-RAG correctly handles inputs that challenge simpler protocols lacking temporal contamination detection or citation tracking.

Input	FreshBench-RAG Output	Baseline Output
Q: Who was appointed as the new CEO of [company] in October 2024? Retrieved: AP article (Oct 14, 2024) naming the new appointee with publication date. Gold span: "[Name], effective November 1, 2024."	Correctly cites the October 14 AP article and outputs the name and start date. Citation precision: 1.0.	GPT-4o-Direct outputs the previous CEO's name from parametric memory (pre-October 2024), citing no passage. Citation precision: 0.0.
Q: What country hosted the emergency G7 summit convened in September 2024? Retrieved: Reuters article (Sep 22, 2024) with location and attendee list. Gold span: "Italy, at an emergency session in Rome."	Correctly retrieves and cites the Reuters passage; outputs "Italy (Rome), September 2024." No stale-answer contamination flagged.	ColBERT+GPT-4o retrieves a 2023 background article on G7 scheduling and outputs "Japan" (the host of the 2023 Hiroshima summit), an unsupported citation error.

Table 12: Success cases: FreshBench-RAG correctly handles these time-sensitive examples while the baseline (top-scoring prior method from Table 1) fails or expresses lower confidence due to reliance on parametric priors or retrieval of temporally misaligned documents.

E.2 Failure Analysis

We identify two recurring failure patterns from the 300-sample error analysis in Section **??**.

Failure Pattern 1: Borderline Stale-Answer Cases. FreshBench-RAG occasionally fails on questions where the ground-truth answer changed within the final 48 hours of the evaluation window, because our rolling index may not yet contain the most recent update. Example: a question about a country's current prime minister was answered correctly by the retrieved BBC article (published 3 days before evaluation), but the prime minister resigned 36 hours before the evaluation timestamp and our index had not yet propagated the new article. FreshBench-RAG output the retrieved (now-outdated) name with citation precision 1.0 relative to the retrieved passage, but F1 of 0.0 relative to the gold span from the newer document. This occurs because indexing latency (typically 24–72 hours for our pipeline) creates a coverage gap for very recent events. Future work could address this by reducing index propagation latency through a streaming ingestion layer (e.g., Kafka-based real-time indexing) or by introducing an explicit "indexing lag" flag that reduces confidence scores for questions whose gold documents are within 72 hours of the evaluation timestamp.

Failure Pattern 2: Distribution Shift on Domain-Specific Named Entities. Performance degrades on questions involving technical acronyms or domain-specific named entities from financial filings and scientific preprints, which are absent from our Reuters/BBC/AP/Wikipedia corpus. Example: a question about a regulatory filing abbreviation (e.g., a specific SEC form number introduced in 2024) was answered incorrectly because no retrieved passage contained the relevant gold span; retrieval returned topically related but terminologically misaligned financial news articles. FreshBench-RAG correctly assigned a low citation-precision score (0.0) to the generated answer, signaling that the retrieval step failed, but could not generate a correct answer from the available context. This is consistent with the dataset coverage limitation discussed in Section **??**.

E.3 Comparison with Best Baseline

Input	FreshBench-RAG	Best Baseline (Hybrid+GPT-4o, no temporal filter)
Q: What verdict was reached in the [high-profile trial] concluded in November 2024? Gold span from AP article (Nov 18, 2024): "Guilty on all counts." Retrieved passages include both the November 2024 verdict article and a March 2024 background article on jury selection.	Correctly identifies the November 2024 AP article as the most temporally relevant passage; outputs "Guilty on all counts" with citation to the November article. F1: 1.0, P_{cite}: 1.0.	Without temporal contamination detection, the baseline's training data contains leaked summaries of early trial coverage. Outputs "acquitted" (matching a pre-verdict media prediction from a contaminated training item). F1: 0.0, P_{cite}: 0.0. This is a stale-answer error driven by pretraining contamination rather than retrieval failure.

Table 13: Direct comparison on a challenging stale-answer example where FreshBench-RAG's temporal contamination detection and citation tracking enable correct handling while the best baseline (Hybrid+GPT-4o without filtering) fails due to a memorized pre-verdict answer overriding the retrieved evidence.